

Practical Session 08: Building & Choosing Financial LLMs

Hands-On: DAPT Budgeting, Hybrid Routing, and a FinBERT vs. LLM Bench

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- 1 Session Overview & Recap
- 2 Guided Problems
- 3 Python Case Study
- 4 Discussion

Duration: ~1 hour (50 min work + 10 min debrief)

Timed agenda

- ➊ **Recap** (8 min): the DAPT objective + the effective-cost formula.
- ➋ **Problem 1** (12 min): size a DAPT corpus and budget for a FinBERT-scale encoder.
- ➌ **Problem 2** (15 min): find the cost-optimal **hybrid routing** fraction ρ for a filing classifier.
- ➍ **Case Study** (15 min): run a FinBERT-style vs. general-model sentiment comparison in Python (code/notebooks/08-domain-specific-llms/).
- ➎ **Discussion** (10 min): choose an architecture for three real briefs.

Recap: the Two Formulas You Will Use

UNDER THE HOOD

DAPT objective. From general weights θ_0 , continue pre-training on a domain corpus $\mathcal{D}_{\text{domain}}$:

$$\theta^* = \arg \min_{\theta} \mathcal{L}_{\text{PT}}(\theta; \mathcal{D}_{\text{domain}}), \quad \mathcal{L}_{\text{PT}} \in \{\text{MLM}, \text{CLM}\},$$

reduced LR ($\sim 10\times$) to limit catastrophic forgetting.

Rule of thumb: encoders need 1–5B domain tokens; the first $\sim 1\text{B}$ yields most of the gain (diminishing returns).

Recap: the Effective-Cost Formula

UNDER THE HOOD

Effective cost per correct output.

$$C_{\text{eff}}(m, \rho) = \frac{c_m + \rho c_h}{p_m + \rho(1 - p_m)},$$

c_m = model cost/query, c_h = human-review cost, p_m = model correctness, ρ = fraction routed to review. Constraint: $p_m + \rho(1 - p_m) \geq p^*$.

How to read it

The numerator is the cost you pay per query (model always, human only on the routed fraction ρ); the denominator is the accuracy you achieve. Routing more to humans raises both — the optimisation finds the smallest ρ that still meets p^* .

Problem 1: Sizing a DAPT Corpus

Scenario. You will DAPT a BERT-base encoder (110M params) for SEC-filing NER, then fine-tune. You can draw from:

- SEC EDGAR (~50B tokens, open)
- Reuters TRC2 (~2B, licensed)
- General replay data (Pile, ~unlimited)

Tasks.

- A. Given diminishing returns, what domain-token budget do you target, and why not just use all 50B?
- B. You worry about catastrophic forgetting. Propose a *replay* ratio (general:domain) and justify it.
- C. Should you set the DAPT learning rate above or below the original pre-training LR? By roughly how much, and what failure does this prevent?
- D. Would TAPT help here? What additional data would it require?

Problem 2: Optimal Hybrid Routing

Scenario. A FinBERT classifier costs $c_m = \$0.0002/\text{query}$ at $p_m = 0.88$. Hard queries route to a human at $c_h = \$2.00/\text{query}$ (assume review fixes all errors). Target accuracy $p^* = 0.95$.

Tasks.

- A. Write the achieved accuracy as a function of ρ and find the *minimum* ρ that meets p^* .
- B. Evaluate C_{eff} at that ρ .
- C. Compare against routing *everything* to the human ($\rho = 1$). What is the cost ratio?
- D. Qualitatively: if p_m rose to 0.95 via DAPT, what happens to the required ρ and to C_{eff} ?

Hint

Achieved accuracy = $p_m + \rho(1 - p_m)$ because review recovers the $(1 - p_m)$ errors on the routed fraction.

Case Study: FinBERT-style vs. General Sentiment

Goal. Demonstrate *polarity inversion* — the single clearest reason finance needs its own models — on a few labelled financial sentences.

Notebook: `code/notebooks/08-domain-specific-llms/demo.ipynb`

Stack:

- `transformers` — load `ProsusAI/finbert` (the Araci FinBERT) and a general sentiment model.
- A handful of finance sentences where general English polarity is misleading.

Prerequisites

`pip install transformers torch`. First run downloads model weights (~440MB for FinBERT). No GPU required for a few sentences.

Case Study: The Code (1/2) — Setup

```
from transformers import pipeline

# Domain-specific encoder (Araci FinBERT) vs. general sentiment
finbert = pipeline("text-classification", model="ProsusAI/finbert")
general = pipeline("sentiment-analysis") # default SST-2 model

SENTENCES = [
    "The firm took on additional liability to fund the acquisition.",
    "We are cautiously optimistic but flag meaningful headwinds.",
    "Leverage increased to 3.2x following the debt issuance.",
    "Unprecedented demand drove revenue above guidance.",
]
```

Two pipelines, four finance sentences chosen so general-English polarity is misleading. The comparison loop is on the next slide.

Case Study: The Code (2/2) — Comparison Loop

```
for s in SENTENCES:
    f = finbert(s)[0]
    g = general(s)[0]
    print(f"{s[:48]:48s} | FinBERT={f['label']:8s} "
          f"| General={g['label']}")
```

Run it, then fill in the table on the next slide.

Case Study: Diagnostic Questions

Q1. For “additional *liability*” and “*leverage* increased”, what does the general model predict vs. FinBERT? Which is right for a *financial reader*, and why does the general model err?

Q2. “*Cautiously optimistic* but ... headwinds” — which label is appropriate? Connect this to the FinBERT-tone earnings-call example from the lecture.

Q3. “*Unprecedented* demand ...” is genuinely positive here. Does FinBERT over-correct? What does this tell you about *context*-dependence of financial polarity?

Q4 — tie to the chapter

Of the four general-model failure modes (polarity inversion, numerical hallucination, table-parsing, regulatory ambiguity), which one does this experiment isolate? Which *cannot* be diagnosed by a sentence-level sentiment test, and why?

Discussion: Pick an Architecture for Each Brief

For each, state the model family + training strategy and justify on accuracy, cost, latency, governance, and traceability.

- ❶ A bank must label **50,000 headlines/day** as material / non-material for insider-trading surveillance. Latency <50ms; 10k labelled examples.
- ❷ A research desk needs **hawkish/dovish** scoring of every new FOMC statement, with the precise monetary-policy vocabulary.
- ❸ A compliance team must answer questions about **any 10-K** in a 2,000-name portfolio, citing the exact source paragraph for audit.

Frame your answer

Encoder vs. decoder? From-scratch vs. DAPT vs. fine-tune-only? Single model vs. hybrid routing vs. RAG? Where does the LM dictionary / register argument bite?

Solution: Problem 1 — DAPT Corpus

A — Budget

Target **~3–5B** EDGAR tokens. Most of the gain is in the first $\sim 1\text{B}$; beyond $\sim 5\text{B}$ the marginal downstream improvement for a 110M encoder is small and not worth the compute or the added risk of forgetting.

B — Replay. Mix in $\sim 5\text{--}10\%$ general (Pile) tokens. Enough to preserve broad language ability, small enough that the domain still dominates the update.

C — Learning rate. *Below* the original PT rate, by $\sim 10\times$. This prevents catastrophic forgetting — a large LR would overwrite the general representations DAPT is meant to *reuse*.

D — TAPT. Yes: a further continued-PT pass on the *unlabelled* text of the NER dataset (same distribution as the labelled set), before supervised fine-tuning. Requires task-distribution unlabelled data, which here we have.

Solution: Problem 2 — Optimal Routing

A — Required ρ . Achieved accuracy $= p_m + \rho(1 - p_m) = 0.88 + 0.12\rho$. Set ≥ 0.95 :

$$0.88 + 0.12\rho \geq 0.95 \Rightarrow \rho \geq \frac{0.07}{0.12} \approx \mathbf{0.583}.$$

B — C_{eff} at $\rho = 0.583$:

$$C_{\text{eff}} = \frac{0.0002 + 0.583 \cdot 2.00}{0.88 + 0.583 \cdot 0.12} = \frac{1.1662}{0.95} \approx \mathbf{\$1.23} \text{ per correct output.}$$

C — All-human ($\rho = 1$): $C_{\text{eff}} = (0.0002 + 2.00)/1.0 \approx \2.00 . Hybrid is $\approx 1.6\times$ cheaper at the *same* 95% target.

D — If DAPT lifts p_m to 0.95. Required $\rho = 0$ (target already met), so

$C_{\text{eff}} \approx c_m / p_m = 0.0002 / 0.95 \approx \0.0002 — four orders of magnitude cheaper. *Investing in the model collapses the human-review bill.*

Wrap-Up: What This Session Showed

- **Problem 1:** DAPT is a budgeting exercise — a few billion domain tokens, a reduced LR, and a replay ratio buy most of the gain. More data is not always better.
- **Problem 2:** hybrid routing meets an accuracy target at a fraction of all-human cost — and the value of *model* improvement (DAPT raising p_m) is dominated by the human-review savings it unlocks.
- **Case study:** polarity inversion is real and visible in four sentences — the cleanest empirical motivation for domain-specific models.
- **Discussion:** architecture choice follows the task's label space, vocabulary sensitivity, latency, and *auditability* requirements.

Next

Financial NLP & sentiment in depth: fine-tuning FinBERT end-to-end and benchmarking it against zero-shot frontier models.

APPENDIX

Stretch Problems

Extra / optional material — beyond the core lecture.

Appendix — Stretch Problem 1: Contamination

A vendor claims their 70B model scores 0.97 F1 on Financial PhraseBank — above any published FinBERT.

Tasks.

- 1 Why is this number *not* immediately trustworthy?
- 2 Design a contamination-controlled evaluation: what temporal property must the test set satisfy relative to the model's knowledge cutoff?
- 3 PhraseBank predates almost every frontier model. What kind of held-out set would you construct instead, and from what source?

Key idea: a benchmark in the pre-training corpus measures memorisation, not generalisation. Require test data postdating the cutoff.

Appendix — Stretch Problem 2: RAG vs. Bigger Model

A 7B decoder + EDGAR RAG vs. a 50B ungrounded decoder, both for “extract the FY revenue from this 10-K”.

Tasks.

- 1 On which axis does RAG win even if the 50B model is slightly more accurate ungrounded? (Hint: SR 11-7.)
- 2 Sketch why grounding reduces *numerical* hallucination specifically.
- 3 Give one task where the 50B model would still be preferable, and say why retrieval does not help there.

Expected: RAG wins on traceability + cost; grounding pins figures to retrieved text; the big model still wins on open-ended reasoning that no single passage supports.